

Mehedi Hasan

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EDUCATION

Jahangirnagar University
MSc. in Computer Science & Engineering

Aug 2023 – Sep 2024
GPA: 3.50/4.0

Jahangirnagar University
BSc. in Computer Science & Engineering

Feb 2018 – July 2023
GPA: 3.47/4.0

EXPERIENCE

Samsung R&D Institute Bangladesh
Software Engineer

Dhaka, Bangladesh
Jan 2024 – Present

Innovation & Research | *Key Field: ML, NLP, Agentic AI, XAI, Wearable devices*

- **Developed an interactive prior-art search agent that refines user queries and detects multi-patent obviousness relationships with explanations**; selected for AI-PoC exhibition and currently being integrated into Samsung Research Bangladesh's in-house patent search workflow to streamline office-wide innovation processes.
- Proposed initial idea and contributed to the development of a **novel machine learning-based methodology for liquid consumption estimation using wearable devices**.
- Developed a weekly report summarizer from GitHub commit messages.
- Fine-tuned Bangla NER model and deployed on a mobile device.
- Actively contribute to team innovation meetings by evaluating peer-submitted ideas, providing constructive technical feedback on novelty, obviousness, and feasibility.

Development | *Tech Stack: JAVA, Kotlin, Jetpack Compose, Android, HILT, WearOS*

- Worked on Samsung Galaxy Wearable projects, developing core modules (notifications, nowbar, watchface overlay), optimizing performance, and prototyping new features for Galaxy Watch 7 and 8.
- Designed and built the complete backend module for the ongoing activity manager (Nowbar) using concurrent data structures
- Developed nowbar overlay module integrating Google's AOD interface and Samsung's new Low Power AOD chip.
- Migrated MVP-based settings module from XML to Jetpack Compose-based MVVM architecture, reducing load time (30ms to 5ms).
- Developed a POC for a link-preview system for the watch notification panel, identifying security vulnerabilities and corresponding solutions.
- Built a backup-and-restore (BnR) module to preserve user settings when changing/connecting new devices.
- Delivered high-quality fixes for 175+ QA/market-reported issues with team's lowest rejection rate.

PUBLICATIONS

- **Mehedi Hasan, Mahbub Islam Mahim, Jugal Krishna Das (2026). Segmentation Strategies for Long-Code Authorship Detection: Balancing Context Coverage and Computational Efficiency.** | *Submitted*
 - Compared truncation, multi-view cropping, and sliding-window segmentation strategies for long-code authorship detection under transformer context limits. Sliding-window achieved the best performance, while multi-view reached near-parity with far fewer computations, highlighting context coverage as the key driver of accuracy.
- **Mehedi Hasan, Mahbub Islam Mahim (2025). Divide and Conquer: A Two-Stage Label-Wise Cascaded Framework for Bangla Hate Speech Detection.** | [Codes](#) [Paper](#)
 - Developed a cascaded pipeline combining binary and multiclass models with k-fold cross-validation and intra-label ensembling; achieved a 0.7392 micro-F1 score, awarded Received **Best Shared Task Paper Award** at the BLP Workshop 2025, and presented at AACL-IJCNLP 2025.

SELECTED PROJECTS

- **Layerwise-Probes-Ablations-in-BERT** – *PyTorch, Hugging Face Transformers, scikit-learn, Matplotlib*
 - Analyzed BERT’s 12-layer encoder using linear probing and layer ablation to distinguish layer-wise responsibilities. Demonstrated that mid-layers (5–6) yield the most separable features, while upper layers (10–11) drive final predictions in parts-of-speech tagging and semantic similarity tasks.
- **Human-Machine-Hybrid-Adversarial-Code-Classification** – *PyTorch, scikit-learn, NumPy, Pandas*
 - Investigated multiple approaches to multilingual code classification — including AST-based feature engineering, overlapping window analysis, multi-view learning, and specialized training loops
- **Image Forgery Detection** – *OpenCV, NumPy, SciPy, scikit-learn, Matplotlib, Seaborn*
 - Implemented copy-move and splicing image forgery detection using LBP, DCT, and SVM, achieving 97% accuracy on CASIA and 89% on COLUMBIA datasets.

COMPETITIONS & ACHIEVEMENTS

Competitive programming & Problems solving

- Solved **1,900+ algorithmic problems** across multiple online judges, participated at **160+ contests**.
- Achieved highest rating of **1611 (Expert)** on Codeforces (@MhD88) and **1730 (3-star)** on CodeChef (@mhd88).
- Top Contest Ranks: Top 2% among 12,275 participants (@MhD88), Top 2% among 16,010 participants (@MhD88)

Hackathons & ML Contests

- Became **Finalist among 400+** teams (Team: JU_NONAME) at Code Samurai Hackathon 2022, hosted by University of Dhaka. Solved algorithmic and SQL problems in preliminaries; developed a web-based system in the final.
- Ranked **1st among 90 teams** (Team: team_hate_mate) at the BLP Workshop 2025 Shared Task

SCHOLARSHIPS & AWARDS

- Received **Excellence Award** twice for contribution in innovation and development at Samsung during Apr-Jun 2025 and Jan-Mar 2024
- Received an award for achieving the professional-level Software Coding (SWC) certification test on November 08, 2024, in recognition of coding excellence.
- Received **Dean’s Merit scholarships** in all four years of B.Sc. for academic excellence (2018–2023)
- Received prestigious national merit scholarship in both primary and secondary school certification examination (2011, 2013)

TEACHING EXPERIENCES

Competitive Programming Trainer | *Department of CSE, Jahangirnagar University* 2022 - 2023

- Provided training on C/C++, Data Structures & Algorithms, problem solving strategies
- Organized weekly programming contests; and coordinated annual sessions to attract new students to community.

Academic Tutoring | *Private Tutor* 2020 - 2024

- Provided academic mentorship to over 80 students across high school, college, and university levels in physics, mathematics, and programming disciplines.

SKILLS

Programming Languages	: C, C++, Java, Kotlin, Python, JavaScript
ML & AI	: scikit-learn, NumPy, OpenCV, PyTorch, Hugging Face Transformers, LLMs
Frameworks & Technologies	: Android, Node.js, Jetpack Compose, HILT, Bootstrap
Databases	: MySQL, MongoDB
Tools	: Git, Perforce, VS Code, Android Studio